THE GEOPHYSICAL INSTITUTE OF ISRAEL המכון הגיאופיסי לישראל
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REPEATED TDEM MEASUREMENTS AT THE NAHAL HEVER SOUTH SITE (LOCATION MAP AND INTERPRETATION RESULTS)

(intermediate report)

909/100/01 (1) **November 2001**

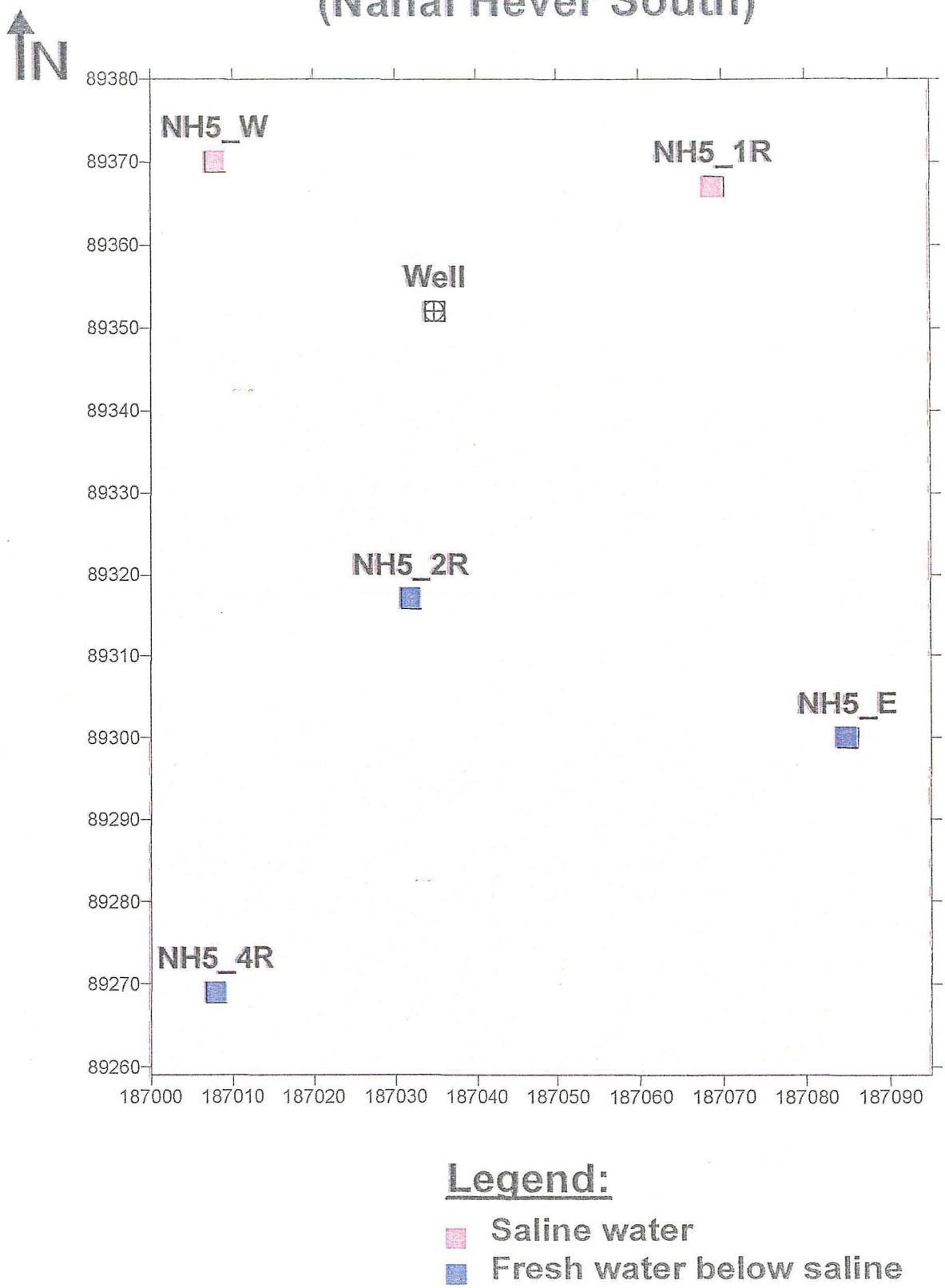
by

Mark Goldman

Prepared for

The Geological Survey of Israel

TDEM location map
(Nahal Hever South)



Legend:

- Saline water
- Fresh water below saline

DATA SET: NH5_1R

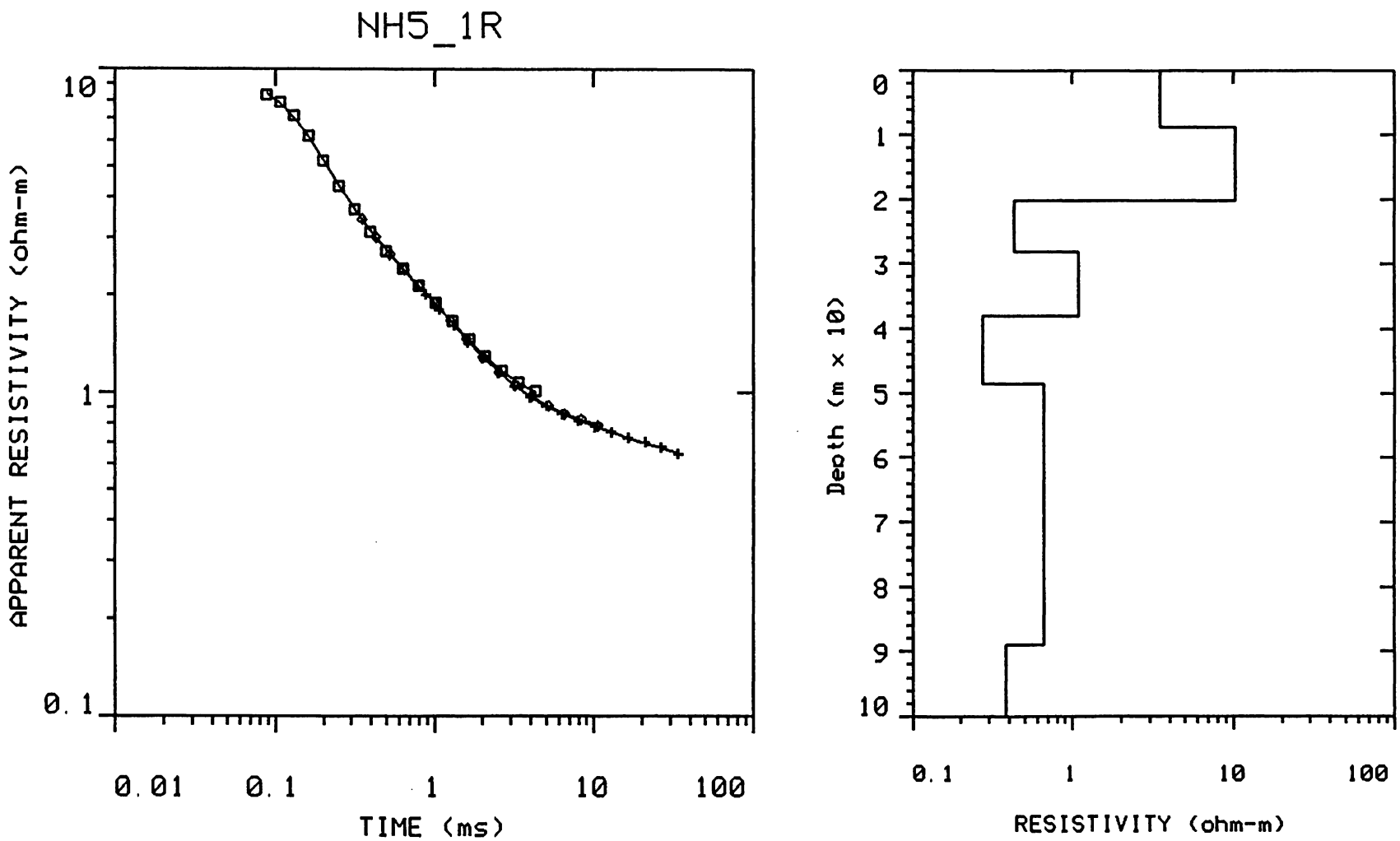
CLIENT: Any Interested Party DATE: 01-DEC-92
LOCATION: Investigation Site SOUNDING: 2
COUNTY: Jefferson County, Colorado ELEVATION: 0.00 m
PROJECT: California Demonstration Data EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 192.0000 N: 220.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 0.898 PERCENT
SMOOTH MODEL FITTING ERROR: 0.956 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			0.0	
1	3.50	8.91	-8.91	2.53
2	10.28	11.32	-20.24	1.10
3	0.434	7.86	-28.10	18.12
4	1.09	9.96	-38.07	9.10
5	0.274	10.44	-48.52	38.05
6	0.662	40.50	-89.02	61.15
7	0.380			

ALL PARAMETERS ARE FREE


NH5_1R
APPARENT RESISTIVITY (ohm-m)
10
1
0.1
0.01
0.1
1
10
100
TIME (ms)
Depth (m x 10)
0
1
2
3
4
5
6
7
8
9
10
0.1
1
10
100
RESISTIVITY (ohm-m)

DATA SET: NH5_W

CLIENT: Any Interested Party DATE: 01-DEC-92
LOCATION: Investigation Site SOUNDING: 2
COUNTY: Jefferson County, Colorado ELEVATION: 0.00 m
PROJECT: California Demonstration Data EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 208.0000 N: 221.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.004 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			0.0	
1	3.91	9.03	-9.03	2.30
2	10.31	10.78	-19.82	1.04
3	0.517	6.79	-26.61	13.12
4	1.01	7.55	-34.17	7.44
5	0.276	9.49	-43.66	34.30
6	0.620	45.43	-89.10	73.22
7	0.411			

ALL PARAMETERS ARE FREE

APPARENT RESISTIVITY (ohm-m)
0.1
1
10
0.01
0.1
1
10
100
TIME (ms)
NH5_W

Depth (m x 10)
10
9
8
7
6
5
4
3
2
1
0
0.1
1
10
100
RESISTIVITY (ohm-m)

DATA SET: NH5_2R

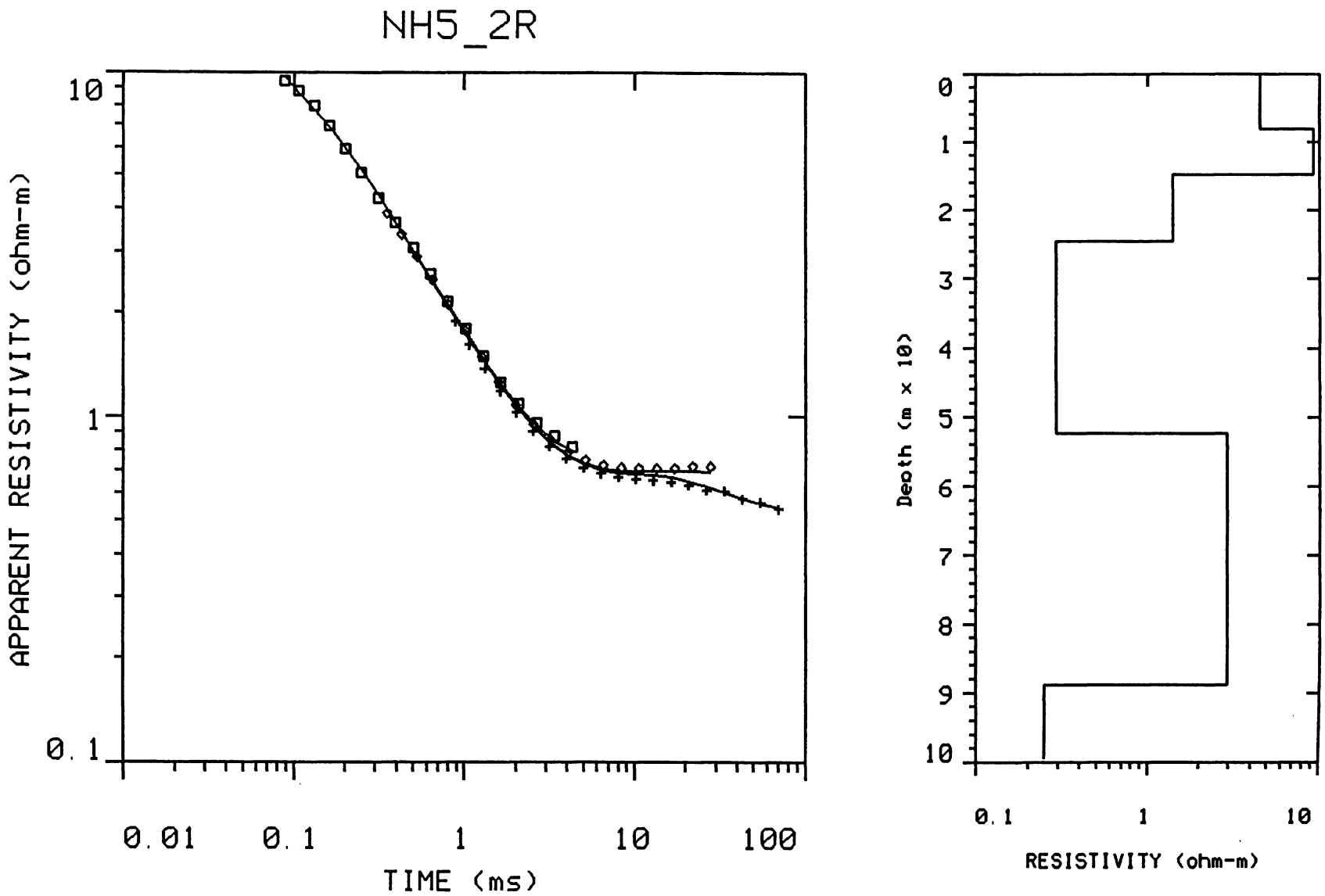
CLIENT: Any Interested Party DATE: 01-DEC-92
LOCATION: Investigation Site SOUNDING: 2
COUNTY: Jefferson County, Colorado ELEVATION: 0.00 m
PROJECT: California Demonstration Data EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 220.0000 N: 222.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.451 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			0.0	
1	4.55	8.19	-8.19	1.79
2	9.30	6.63	-14.82	0.712
3	1.39	9.79	-24.61	7.00
4	0.294	27.76	-52.37	94.39
5	2.88	36.48	-88.86	12.62
6	0.248			

ALL PARAMETERS ARE FREE


NH5_2R
APPARENT RESISTIVITY (ohm-m)
10
1
0.1
0.01
0.1
1
10
100
TIME (ms)
Depth (m x 10)
0
1
2
3
4
5
6
7
8
9
10
0.1
1
10
RESISTIVITY (ohm-m)

DATA SET: NH5_4R

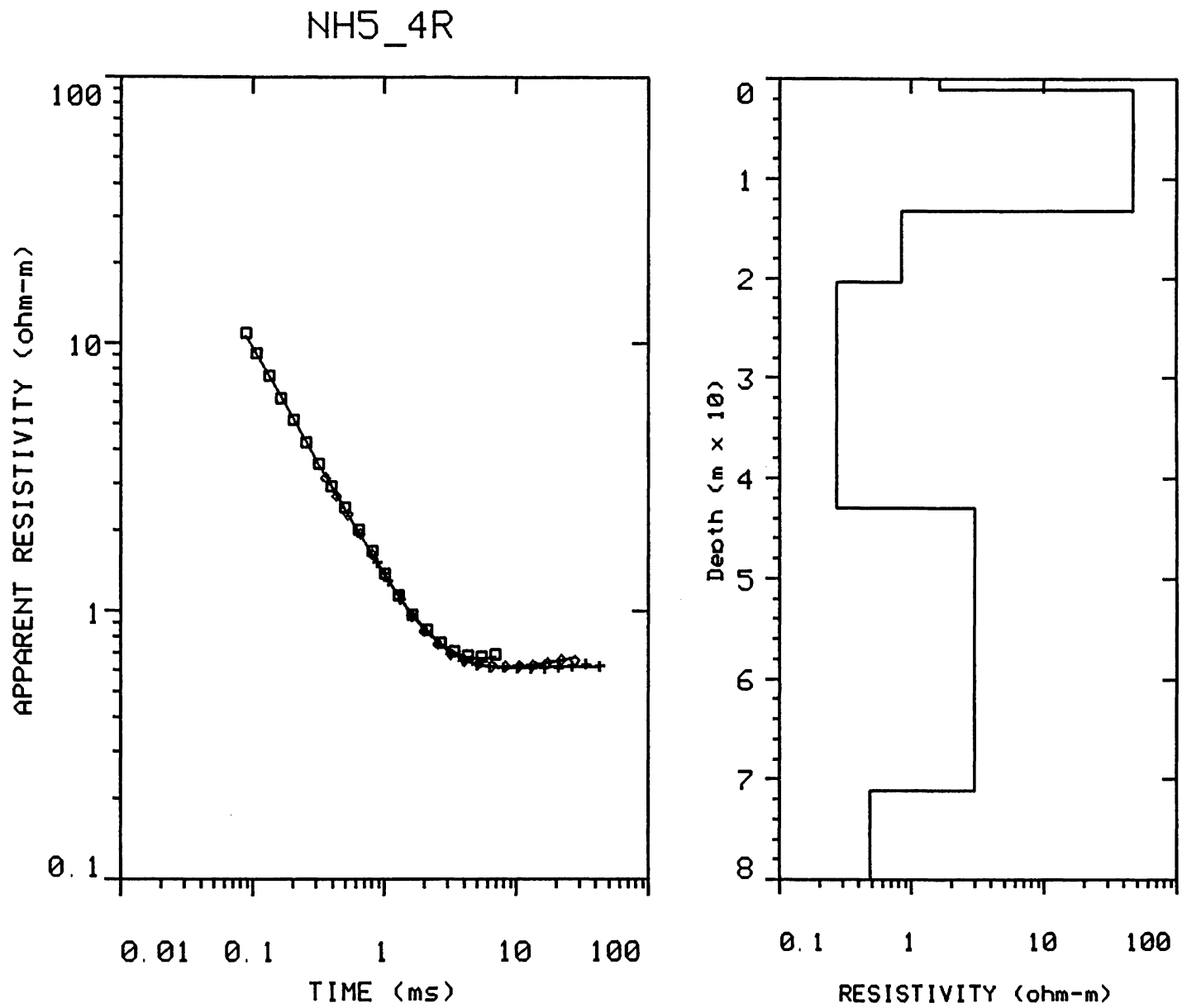
CLIENT: Any Interested Party DATE: 01-DEC-92
LOCATION: Investigation Site SOUNDING: 2
COUNTY: Jefferson County, Colorado ELEVATION: 0.00 m
PROJECT: California Demonstration Data EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 232.0000 N: 223.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.703 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			0.0	
1	1.64	1.09	-1.09	0.664
2	46.68	12.12	-13.21	0.259
3	0.843	7.16	-20.38	8.49
4	0.270	22.63	-43.02	83.73
5	3.00	28.13	-71.15	9.37
6	0.482			

ALL PARAMETERS ARE FREE


NH5_4R
APPARENT RESISTIVITY (ohm-m)
100
10
1
0.1
0.01
0.1
1
10
100
TIME (ms)
Depth (m x 10)
0
1
2
3
4
5
6
7
8
0.1
1
10
100
RESISTIVITY (ohm-m)

DATA SET: NH5_E

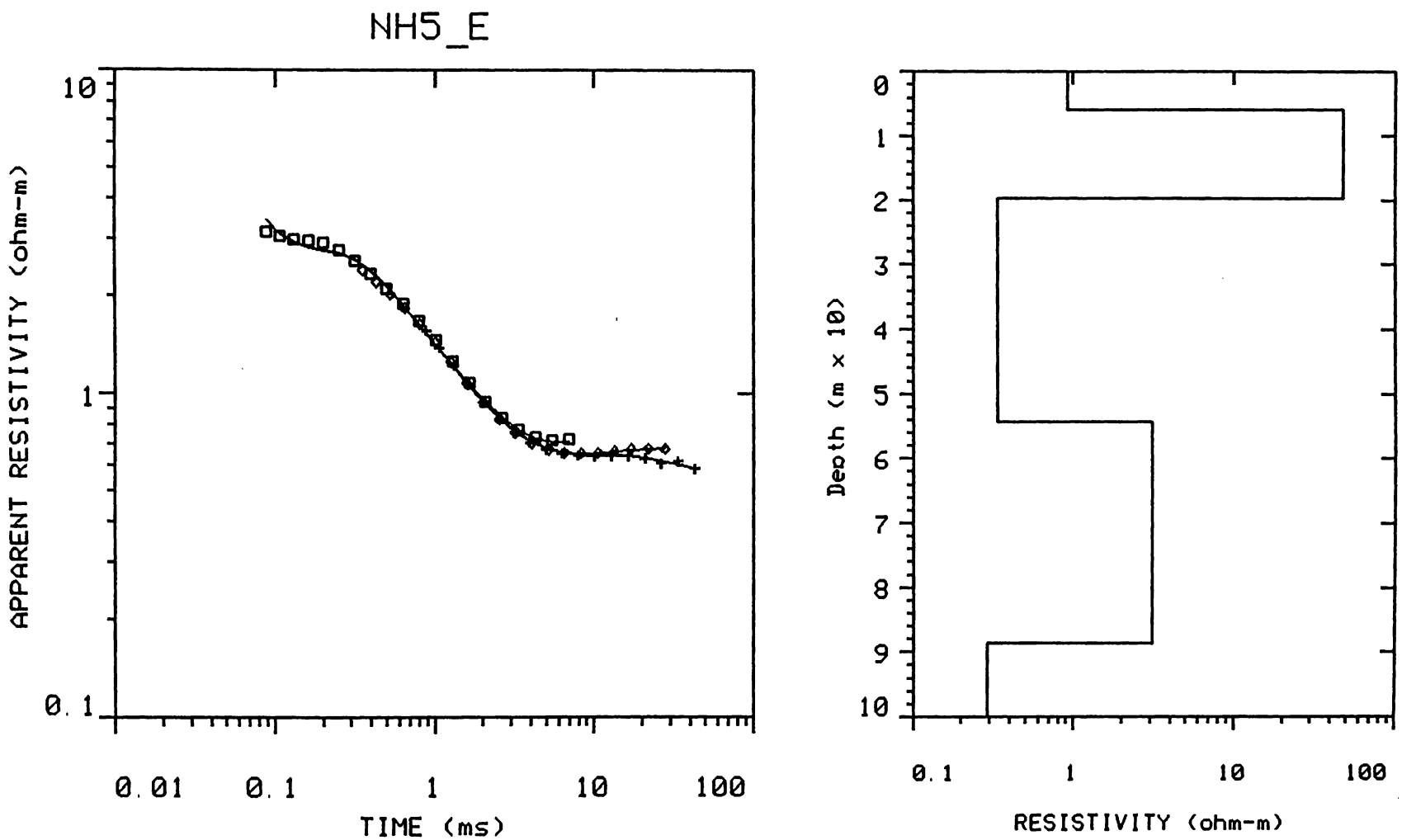
CLIENT: Any Interested Party DATE: 01-DEC-92
LOCATION: Investigation Site SOUNDING: 2
COUNTY: Jefferson County, Colorado ELEVATION: 0.00 m
PROJECT: California Demonstration Data EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 246.0000 N: 224.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.247 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			0.0	
1	0.927	5.82	-5.82	6.28
2	47.53	13.93	-19.76	0.293
3	0.336	34.56	-54.32	102.6
4	3.10	34.22	-88.54	11.01
5	0.292			

ALL PARAMETERS ARE FREE


NH5_E
APPARENT RESISTIVITY (ohm-m)
10
1
0.1
0.01
0.1
1
10
100
TIME (ms)
Depth (m x 10)
0
1
2
3
4
5
6
7
8
9
10
0.1
1
10
100
RESISTIVITY (ohm-m)